

EECE 2560: Fundamentals of Engineering Algorithms

Department of Electrical and Computer Engineering

Project Word Search

Write a program that solves a word search puzzle. The program reads an $n \times n$ grid of letters from a file and the list of k words from the file **dictionary**. The program prints out all the candidate words that can be found in the dictionary. Candidate words can be found in the grid by starting from any letter and reading left, right, up, down, or along any of the four diagonals. Words can also wrap around the edges of the grid. Words must be at least 5 characters long. The output file of each puzzle should list all words which can be found in the dictionary.

The goal is to find an algorithm that solves this problem that runs as quickly as possible for large n and k . Several sample word search puzzles are also provided.

Part a

1. Implement a class called **dictionary** that reads the words from the dictionary file and stores them in a vector, and which includes:
 - (a) a function to read the words from the dictionary file,
 - (b) an overloaded output operator to print the word list,
 - (c) a function that sorts the words using `selectionsort`,
 - (d) a function to handle word lookups using binary search.
2. Implement a class called **grid** that reads the letters in the grid from a file and stores them in a matrix.
3. Implement a global function **findMatches()** that is passed the dictionary and the grid as parameters and prints out candidate words that can be found in the dictionary. That is, scan the grid to get all possible words and search each possible word (at least 5 characters long) in the dictionary. If found, print to an output file with the following format.

word	start point (i, j)	found_index
north	(0,2)	1200

4. Implement a global function **search()** which (1) reads the name of the grid file from the keyboard, (2) read the data from input files for grid and dictionary, (3) sort all words in the dictionary, and (4) prints out candidate words that can be found in the dictionary.

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n y d m k u a s l m o q y r c
u o t e u i t n m o o t w w p
e m r w t u h i d t n r m p h
g s b t d a t q k i r a a y o
d f q e h r c h f v i m u v i
d g n e m e i u b a v s p c l
q t j r q q a w d t p s b a j
s b h y s f u s o e a r e r e
o o r e n m j f t d d n s a p
y e j n a c w j o e k n b w p
v n f a k m k n c c r v r p c
d e t n e l a t r c u n k i q
z s c k q c d c n y l o t g n
s p a q n a w c g s f c i l h
h x p p i z u t w x b g m r a